👌 XAI Lecture 14

Network Dissection & TCAV

FAMAF - UNC

First Semester 2026

eXplainable
Artificial Intelligence

🧉 Disclaimer ---

This course is based on

Explainable Artificial Intelligence

From Simple Predictors to Complex Generative Models

Spring 2023, Harvard University

https://interpretable-ml-class.github.io/

🧉 Paper 1 ---

Network Dissection: Quantifying Interpretability of Deep Visual Representations

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Presented by: Anat Kleiman, Gustaf Ahdritz, Xin Tang, Luke Bailey

[@bau2017network]

🧉 Presentation Roadmap ---

- **Introduction**
  - Motivation
  - Questions the paper aims to answer
  - Related works
- **Method**
- **Experiments**
  - Training Conditions
  - Discrimination
  - Layer Width

🧉 How Is Semantic Visual Concept Represented in the Brain? ---

🧉 A Neuron That Only Fires for Jennifer Aniston ---

🧉 Disentangled Representation in Visual Cortex ---

🧉 Deep CNN for Computer Vision ---

Proposed Questions ---

1. **What is a disentangled representation, and how can its factors be quantified and detected (in deep CNN)?**
2. **Do interpretable hidden units reflect a special alignment of feature space, or are interpretations a chimera?**
3. **What conditions in state-of-the-art training lead to representations with greater or lesser entanglement?**

🧉 Proposed Questions and Contributions ---

1. **What is a disentangled representation, and how can its factors be quantified and detected?**
   - Proposed a metric, intersection over union score (IoU), to quantify the interpretability of each unit
   - The alignment level between unit activated area and human-interpretable concepts
2. **Do interpretable hidden units reflect a special alignment of feature space, or are interpretations a chimera?**
   - A semantic concept can be detected by many units
   - A unit can detect many semantic concepts
3. **What conditions in state-of-the-art training lead to representations with greater or lesser entanglement?**
   - Number of unique detectors, layer depth, training iterations
   - The angle of the images, input datasets
   - Fine-tuning, supervised vs. unsupervised

🧉 Related Works ---

1. **Generative Visualizations of Individual Units**
   - Mahendran et al., CVPR 2015
   - Nguyen et al., NIPS 2016
   - Simonyan et al., ICML 2014
2. **Salience-based Visualizations of Individual Units**
   - Deconvolution: Zeiler et al., ECCV 2014
3. **Visualizing Representations as a Whole**
   - t-SNE: Maaten et al., JMLR, 2008
   - prototype autoencoder: Li et al., AAAI, 2018
   - Yosinski et al., ICML, 2015

Limitation: qualitative analyses, cannot be used for comparison between models.

🧉 Method ---

Broden: Broadly and Densely Labeled Dataset ---

- Combination of multiple datasets with segmentation and image-wide labels

🧉 Scoring Unit Interpretability ---

Unit is a convolutional filter

🧉 Scoring Unit Interpretability ---

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🧉 Scoring Unit Interpretability ---

$$IoU_{k,c} = \frac{\sum |M_k(\mathbf{x}) \cap L_c(\mathbf{x})|}{\sum |M_k(\mathbf{x}) \cup L_c(\mathbf{x})|}$$

- Changing *IoU* threshold changes number of concept detectors but not orderings between networks
- One unit might be a detector for multiple concepts; they choose the top-ranked concept for an individual unit
- Interpretability of a layer = number of unique concepts aligned with units

🧉 Experiments ---

🧉 Experiments: Tested CNN Models ---

Training	Network	Data set or task
none	AlexNet	random
Supervised	AlexNet	ImageNet, Places205, Places365, Hybrid
Supervised	GoogLeNet	ImageNet, Places205, Places365
Supervised	VGG-16	ImageNet, Places205, Places365, Hybrid
Supervised	ResNet-152	ImageNet, Places365
Self	AlexNet	context, puzzle, egomotion, tracking, moving, videoorder, audio, crosschannel, colorization, objectcentric

Places: scene-centric dataset with categories such as kitchen, living room, coast.

🧉 Experiment 1: Human Evaluation of Interpretations ---

1. **Identify Interpretable Units** — units that raters agreed with ground-truth interpretations
2. Raters shown 15 images with highlighted patches showing the most highly-activating regions for each unit in AlexNet trained on Places205, and asked to decide (yes/no) whether a given phrase describes most of the image patches
3. **Find Network Dissection** — portion of interpretations generated by method that were rated as descriptive
4. **Human Consistency** — portion of ground-truth labels found descriptive by a second group of raters

🧉 Experiment 1: Human Evaluation Results ---

	conv1	conv2	conv3	conv4	conv5
Interpretable units	57/96	126/256	247/384	258/384	194/256
Human consistency	82%	76%	83%	82%	91%
Network Dissection	37%	56%	54%	59%	71%

Network Dissection agreement increases in higher layers; Human consistency remains high throughout.

🧉 Experiment 2: Axis-Aligned Interpretability ---

Two hypotheses:

1. **Concepts appear in every direction**
   - **Default hypothesis:** single units not much more interpretable than combinations of units
2. **Concepts are rare + the model converges to a special, semantically rich basis**
   - The model's natural basis is a meaningful decomposition

🧉 Experiment 2: Axis-Aligned Interpretability ---

🧉 Experiment 3: Concepts by Layer ---

🧉 Experiment 4: Network Architectures ---

🧉 Experiment 5: Training Conditions ---

Varied training conditions:

1. **Weight Initializations**
   - **Minimal Effect:** Models converge to similar levels of interpretability
2. **Dropout**
   - **Some Effect:** Lack of dropout leads to more "texture" and less "object" detectors
3. **Batch Normalization**
   - **Significant Effect:** Interpretability decreased significantly

🧉 Experiment 5: Training Conditions ---

🧉 Experiment 6: Discrimination ---

Benchmark high-level activations on a new task:

- Across several Deep NNs, extract activations from high CNN layers
- Train a linear SVM on a new *action recognition* task
- Compute classification accuracy

Experiment 6: Discrimination Results ---

🧉 Experiment 7: Width ---

Effect of layer width (number of units in a layer):

Increased layer width retains similar accuracy, but many more **concept detectors**

- # Detectors increased both at the increased layer and in the network generally
- Increase has a threshold

Experiment 7: Width Results ---

🧉 Discussion Questions (Paper 1) ---

1. **Distribution Understanding:** Concept detectors from a particular dataset betray something about the underlying distribution. How do you think this can be applied in the real world (e.g., bias detection)?
2. **Single unit to circuit:** This method interprets single units; could it be extended to larger circuits and would this be useful?
3. **Beyond vision:** This method is deeply tied to the vision domain. Could it be extended to other domains such as natural language?

Interpretability Beyond Feature Attribution: Quantitative Testing with Concept Activation Vectors (TCAV)

Authors: Kim et al. (2018)

Presented by: Lucia Gordon, Matthew Nazari, Catherine Yeh

[@kim2018interpretability]

🧉 Thoughts on Concept-Based Explanations So Far? ---

Thoughts on **Concept-Based** Explanations So Far?

🧉 Motivation + Problem Statement ---

- Interpreting deep learning models is crucial to understanding their behavior, ensuring accurate predictions, and reflecting our values
- But remains a big challenge due to size, complexity, and opacity of ML models
- Many systems operate on **low-level features** (e.g., pixel values) rather than **high-level concepts** (e.g., face) that are human-interpretable

🧉 Motivation + Problem Statement ---

Problem: - We can't express these concepts as pixels - And they weren't our input features

Summary of Contributions ---

- Introduce **Concept Activation Vectors (CAVs)**: way to interpret a neural network's internal state in terms of human-friendly concepts
- Key idea is to use the high-dimensional internal state of a neural net as an aid, not an obstacle
- Main contribution: **Testing with CAV (TCAV)**, that quantifies model sensitivity to a high-level concept learned by a CAV for a particular class

🧉 Goals of TCAV ---

- **Accessibility:** requires little to no ML expertise
- **Customization:** adaptable to any concept, even outside of training
- **Plug-in readiness:** works without retraining/modifying ML models
- **Global quantification:** can interpret entire classes with a single quantitative measure

$\downarrow$

Assessed with experiments + human evaluation

Related Work: Interpretability Methods ---

Interpretability methods:

- *Inherently interpretable* models vs. *post-hoc* explanations (Kim et al., 2014; Doshi-Velez et al., 2015; Goodman & Flaxman, 2016)
- **Perturbation-based methods:** e.g., LIME/SHAP (Ribeiro et al., 2016; Lundberg & Lee, 2017)
  - *local* vs. *global*

Related Work: Saliency Limitations ---

Interpretability methods in neural networks

- Limitations of **saliency methods**:
  - Local explanation (Erhan et al., 2009; Smilkov et al., 2017)
  - Lack customization
  - Vulnerable to adversarial attacks (Ghorbani et al., 2017)
  - Insensitivity to randomization (Adebayo et al., 2018)

Related Work: Linearity + Latent Dimensions ---

Linearity in neural network + latent dimensions

- Meaningful information can be learned from simple *linear* classifiers (Bau et al., 2017; Alain & Bengio, 2016)
- Mapping *latent* dimensions to human *concepts* (Mikolov et al., 2013; Zhu et al., 2017)

🧉 Approach: Defining the CAV ---

Consider the fully connected layer $f_l : \mathbb{R}^n \to \mathbb{R}^m$ and the concept of interest C

- Collect a set of examples P_C of that concept and a negative set N of examples that don't
- Define the CAV to be a vector orthogonal to a decision boundary between activations $\{f_l(\mathbf{x}) : \mathbf{x} \in P_C\}$ and $\{f_l(\mathbf{x}) : \mathbf{x} \in N\}$

🧉 Approach: Visualizing the CAV ---

The CAV $\mathbf{v}_C^l$ is the normal to the linear decision boundary separating:
- $\in \{f_l(x) : x \in P_C\}$ (concept examples) - $\in \{f_l(x) : x \in N\}$ (non-examples)

🧉 Approach: Gauging "Concept Sensitivity" ---

Saliency maps gauge sensitivity of $h_k(\mathbf{x})$ with respect to per-pixel perturbations:

$$S_{C,k,l}(\mathbf{x}) = \nabla h_{l,k}(f_l(\mathbf{x})) \cdot \mathbf{v}_C^l$$

With CAV, we can gauge sensitivity of $h_{l,k}(f_l(\mathbf{x}))$ **towards a concept** at an entire layer.

🧉 Approach: Testing with CAV (TCAV) ---

$$\text{TCAV}_{Q_{C,k,l}} = \frac{|\{\mathbf{x} \in X_k : S_{C,k,l}(\mathbf{x}) > 0\}|}{|X_k|}$$

- $\text{TCAV}_{Q_{C,k,l}}$ measures the fraction of inputs whose activations were influenced by a concept
- This provides interpretation global to a particular class
- A t-test can safeguard against meaningless CAVs

🧉 Approach Summary ---

🧉 Results: Sorting Images with CAVs ---

🧉 Results: Gaining Insights with TCAV ---

- Matches intuition: "Red" important for "fire engine"; "Striped" important for "zebra" - **Biases:** "Caucasian" important for "rugby ball" - Statistical significance test successfully removed spurious CAVs ("Dotted" not important for "zebra")

🧉 Results: TCAV for Where Concepts Are Learned ---

- Simple concepts (colors, patterns) reach high accuracy at **low layers** - Complex concepts (age, sex, objects) don't reach high accuracy until **higher layers** - Confirming past findings: lower layers = feature detectors, higher layers = classifiers

🧉 Results: Controlled Experiment with Ground Truth ---

Results: Evaluation of Saliency Maps ---

🧉 Results: Saliency Maps vs. Human Subjects ---

🧉 Results: TCAV for a Medical Application ---

🧉 Conclusions ---

- **Roadmap:** Gradient → Attention → Concept based approaches (**TCAV!**) - **Limitations:** - Evaluated only on computer vision tasks - Statistical significance testing — is this rigorous?

🧉 Discussion Questions (Paper 2) ---

- How does this method compare to e.g., saliency maps?
- How would you adapt TCAV for other types of data (e.g., audio/video) and what would that look like?
- Thoughts on TCAV for adversarial example identification (Appendix A)?
- Do you think there could be adversarial images that allow meaningless CAVs to pass the statistical significance test?

Thank You!

🧉 References --- |